Time-of-flight (TOF) determination of the speed of light using short LED light pulses

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Goal

We configure an experiment that measures the speed of light through air by generating a pulse of light and measuring the time it takes for the pulse to traverse a known distance.

Introduction and Background

why do we care about c?
why is it important to science?
cite

Light can be conceptualized as particles that travel across space at a speed. Given the distance that an object travels and the time that the object takes to traverse that distance, we can obtain the average speed of the object in that interval of time and space.

The average speed of an object that traverses a distance d in a time t is

$$v = \frac{d}{t} \quad (1)$$

Assuming the medium does not change, it is understood that the speed of a light pulse is constant. Under this assumption, measuring the average speed of a light pulse provides the instantaneous speed of the light pulse at every point in an interval of time.

~~Additional Equations~~

talk about why you need these

Uncertainty propagation formula, used to estimate uncertainties in derived values:

cite where these came from

$$\delta f^2 = \sum_i \left(\partial_{x_i} f \cdot \delta x_i\right)^2 \quad (2)$$

- δf: estimated uncertainty in f, a function of various inputs x_i
- δx_i: uncertainty in input x_i
- x_i: input i to function f

in sentence form

Mean of a set of a set of values:

$$\bar{x} = \frac{1}{N} \sum_i x_i \quad (3)$$

- $\bar{x}$: mean of the values
- N: number of data points

sentences

- x_i: data point i

Sample standard deviation of a set of values:

$$s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \quad (4)$$

- s: sample standard deviation of values
- $\bar{x}$: mean of values
- x_i: data point i

Sentences

Standard deviation of the mean (SDOM):

$$s_{\bar{x}} = \frac{s}{\sqrt{N}} \quad (5)$$

- s: sample standard deviation
- N: number of samples

sentence form

Procedures and Data

We create a pulse of light using an LED light source ("emitter"). On a timer, this emitter produces short pulses of light at a frequency of 30 kHz. The pulse of light is oriented toward a retroreflector, which reflects the pulse and returns it to the emitter for detection.

The light source includes sensors for detecting the beam's exit and re-entry, measuring the beam intensity at both events. Beam intensities for each signal as a function of time are collected using an oscilloscope, ~~which the light emitter interfaces with using two cables for the input and output signals and one cable for oscilloscope-emitter clock synchronization.~~

The exit and entry intensity signals collected by the emitter produce stable "peaks" of light intensity in the oscilloscope display, separated by an interval of time t_i. This interval of time is considered the time taken by the pulse of light to return to the light source after being emitted and is produced by subtracting the time of emission $t_{e,i}$ ~~(measured from the first peak~~) from the time of return $t_{r,i}$ ~~(measured from the second peak~~). When performing readings, we collect upper and lower bounds for these times and consider the uncertainty of time measurements to be half of the interval from the lower bound to the upper bound.

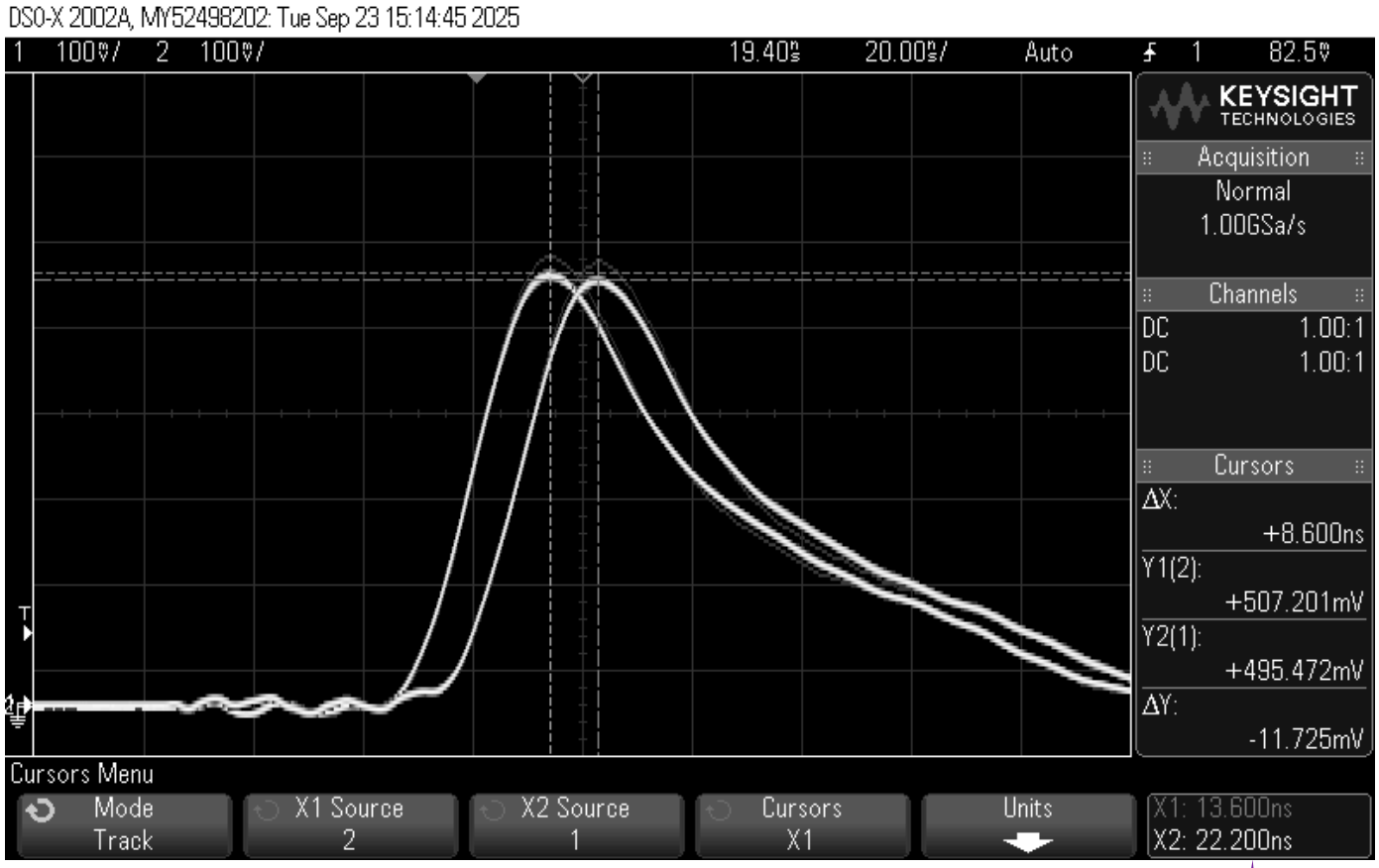


Figure 1 - Intensity peaks as viewed on oscilloscope Be more descriptive

To mitigate loss of beam intensity, we use a Fresnel lens to produce parallel light beams from the initially divergent light beams exiting the emitter. Parameters of the lens configuration, including distance from the light and angles of orientation, are adjusted so as to maximize the intensity of the returning beam visible on the oscilloscope.

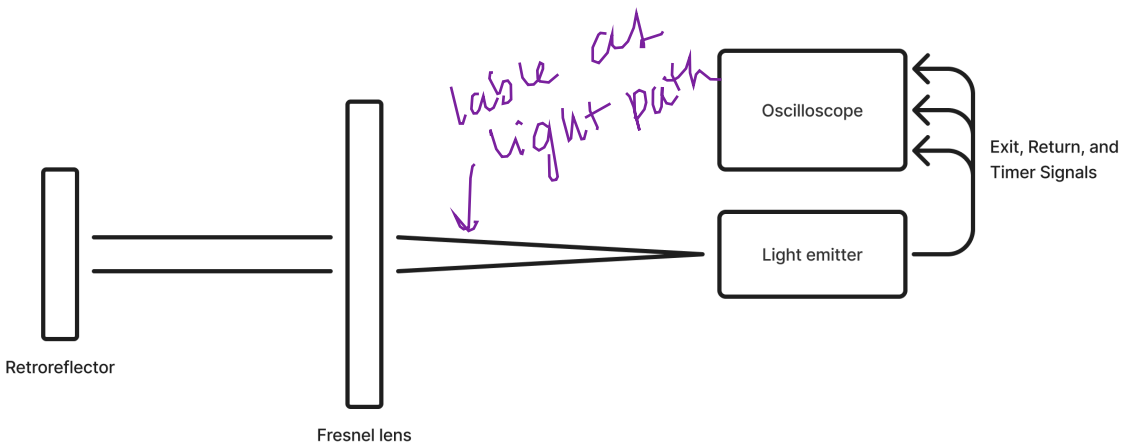


Figure 2 - Experimental apparatus Be more descriptive

We position the reflector at various distances, measured as d_i, from the emitter. For each configuration, we use a tape measure to measure the distance from the emitter to the reflector and adjust the Fresnel lens as described above.

give range of distances

need uncertainties!!

Configuration	d_i - Half of distance (cm)	Time of exit, lower bound (ns)	Time of exit, upper bound (ns)	Time of return, lower bound (ns)	Time of return, upper bound (ns)
0	155.5	12.6	13.8	21.0	22.6
1	175.5	11.8	13.4	21.6	23.0
2	195.5	11.0	12.6	22.0	24.8
3	215.5	10.2	11.6	23.0	24.8
4	235.5	7.8	10.6	23.2	24.6
5	255.5	7.2	10.0	22.8	25.2
6	275.5	6.0	8.0	23.6	25.6
7	295.5	5.6	7.2	24.2	26.4
8	315.5	3.0	5.2	23.0	25.2

Table 1 : give description

Note that the distances above are half of the distance the light traversed in the corresponding time interval measured. To obtain a speed from a single data point, one might compute $c_i = 2d_i/t_i$.

Can go in analysis

Manipulating the average speed expression (using $2d$ as the distance):

$$v = \frac{2d}{t} \Rightarrow 2d = v \cdot t$$

This suggests that we expect a positive linear association between the distance traveled by the light pulse and the time taken for the pulse to travel said distance, where the sensitivity of the distance traveled to the time taken is the speed of the light pulse. We can use least-squares regression to estimate this relationship; the slope of the resulting line of best fit approximates the speed of the light pulse.

~~Uncertainty Considerations~~

in analysis

In addition to the treatment of uncertainty when reading time intervals from the oscilloscope, we also consider uncertainty introduced when measuring distances. The tape measure used for measuring distances has a metal tip, which is approximately 1mm in width. The resolution of the tape measure is approximately 1mm. Despite attempting to align the retroreflector as closely as possible with our reference point, we estimate that slight deviations in angle with the plane orthogonal to the light's central axis contribute 3mm of uncertainty to distance measurements.

We note other sources of length measurement uncertainty that introduce a uniform length shift to our data points. These include the width of the retroreflector and the embedding depth of the emitter exit and re-entry sensors. We may neglect these in a slope analysis, as they contribute only to the predicted intercept of a linear regression model.

Results and Analysis

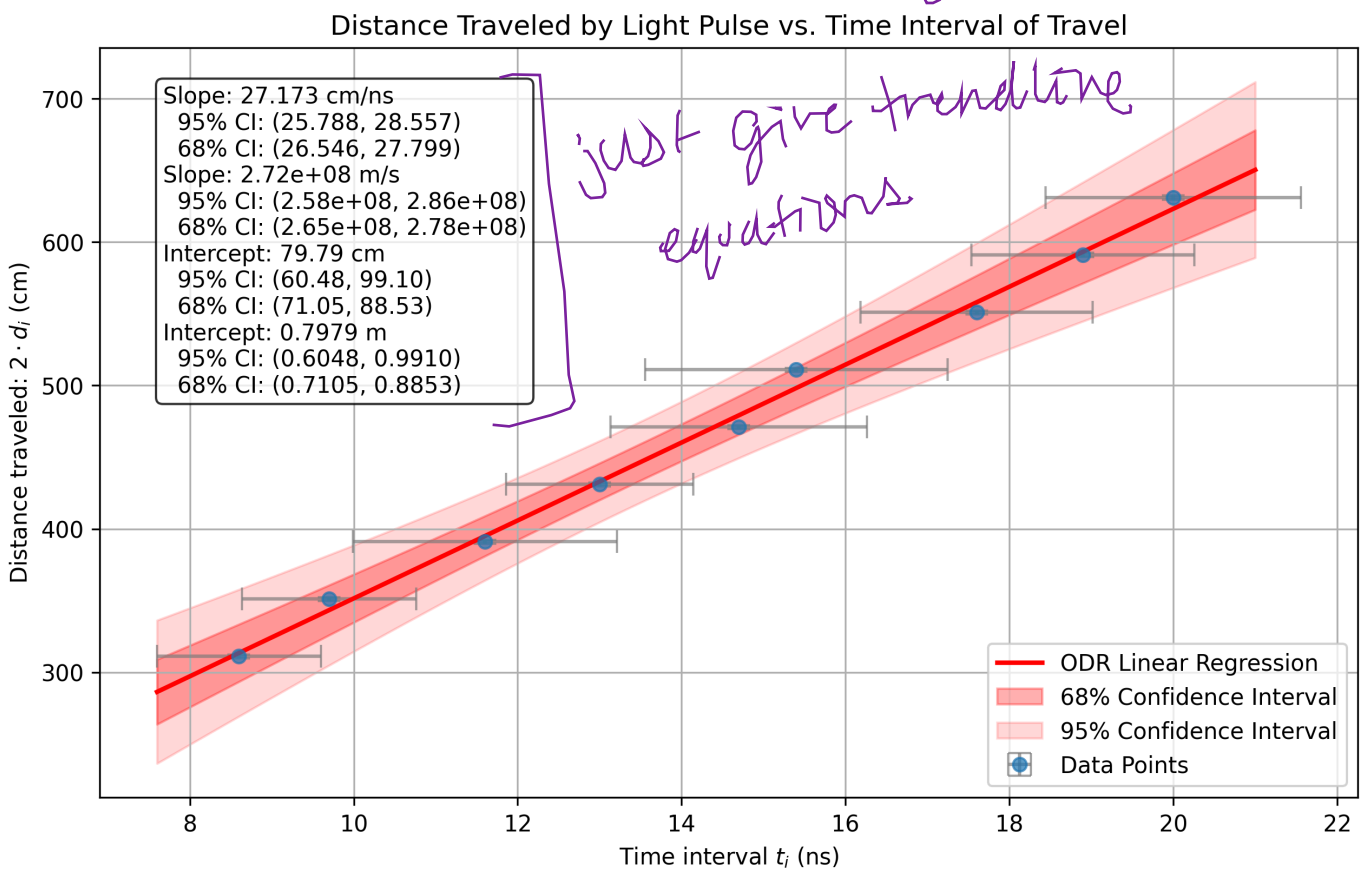
Linear Regression

We approximate a line of best fit to our dataset using orthogonal least squares and obtain a slope of approximately:

$$(27.172 \pm 1.3845)\frac{\text{cm}}{\text{ns}} = (2.72 \cdot 10^8 \pm 1.3845 \times 10^7)\frac{\text{m}}{\text{s}}$$

to 68% confidence

Sig Figs



just give trendline equations

Figure 3 - Linear regression analysis

Be more description

Pointwise Analysis

Using individual points to compute the speed of light with the equation $c_i = 2d_i/t_i$ we obtain one measurement of the speed of light per apparatus configuration:

you have use this multiple times just call it Equation + reference

	Speed of pulse - c_i
0	36.1628 ± 4.20658
1	36.1856 ± 3.96688
2	33.7069 ± 4.68620
3	33.1538 ± 2.90880
4	32.0408 ± 3.41237
5	33.1818 ± 3.97353

Sig Figs!

	Speed of pulse - c_i
6	31.3068 ± 2.51624
7	31.2698 ± 2.25097
8	31.5500 ± 2.45452

Table 2: description

For point 0, this computation is:

don't need this

$$
\begin{aligned}
\delta t_e &= \frac{1}{2}(t_{e,\text{upper}} - t_{e,\text{lower}}) = \frac{1}{2}(13.8\text{ns} - 12.6\text{ns}) = 0.6\text{ns} \\
\delta t_r &= \frac{1}{2}(t_{r,\text{upper}} - t_{r,\text{lower}}) = \frac{1}{2}(22.6\text{ns} - 21.0\text{ns}) = 0.8\text{ns} \\
\delta t &= \sqrt{(\partial_{t_e} t \cdot \delta t_e)^2 + (\partial_{t_r} t \cdot \delta t_r)^2} \\
&= \sqrt{1\delta t_e^2 + 1\delta t_r^2} = \sqrt{(0.6\text{ns})^2 + (0.8\text{ns}^2)} = 1.0\text{ns} \\
t_e &= \frac{1}{2}(t_{e,\text{upper}} + t_{e,\text{lower}}) = \frac{1}{2}(13.8\text{ns} + 12.6\text{ns}) = 13.2\text{ns} \\
t_r &= \frac{1}{2}(t_{r,\text{upper}} + t_{r,\text{lower}}) = \frac{1}{2}(22.6\text{ns} + 21.0\text{ns}) = 21.8\text{ns} \\
t &= t_r - t_e = 8.6\text{ns} \\
\delta d &= 0.5\text{cm} \\
d &= 155.5\text{cm} \\
c &= \frac{2d}{t} = \frac{2 \cdot 155.5\text{cm}}{21.8\text{ns}} = 36.1628\frac{\text{cm}}{\text{ns}} \\
\delta c &= \sqrt{(\partial_d c \cdot \delta d)^2 + (\partial_t c \cdot \delta t)^2} = \sqrt{\left(\frac{2}{t}\delta d\right)^2 + \left(\frac{-2d}{t^2}\delta t\right)^2} \\
&= \sqrt{\left(\frac{2}{8.6\text{ns}} \cdot 0.5\text{cm}\right)^2 + \left(\frac{2 \cdot 155.5\text{cm}}{(8.6\text{ns})^2} \cdot 1\text{ns}\right)^2} \\
&= 4.207\frac{\text{cm}}{\text{ns}}
\end{aligned}
$$

We obtain the sample mean, standard deviation, and standard deviation of the mean from our dataset:

- Sample mean (equation 3): $33.173\ \text{cm} / \text{ns}$
- Sample standard deviation (equation 4): $1.914\text{cm} / \text{ns}$
- Standard deviation of the mean (equation 5): $0.638\ \text{cm} / \text{ns}$

in sentence form

and obtain a value of the speed of light:

$$33.173\frac{\text{cm}}{\text{ns}} \pm 0.638\frac{\text{cm}}{\text{ns}} = (3.3173 \times 10^8)\ \frac{\text{m}}{\text{s}} \quad \pm \quad (6.38 \times 10^6)\ \frac{\text{m}}{\text{s}}$$

Sig figs!

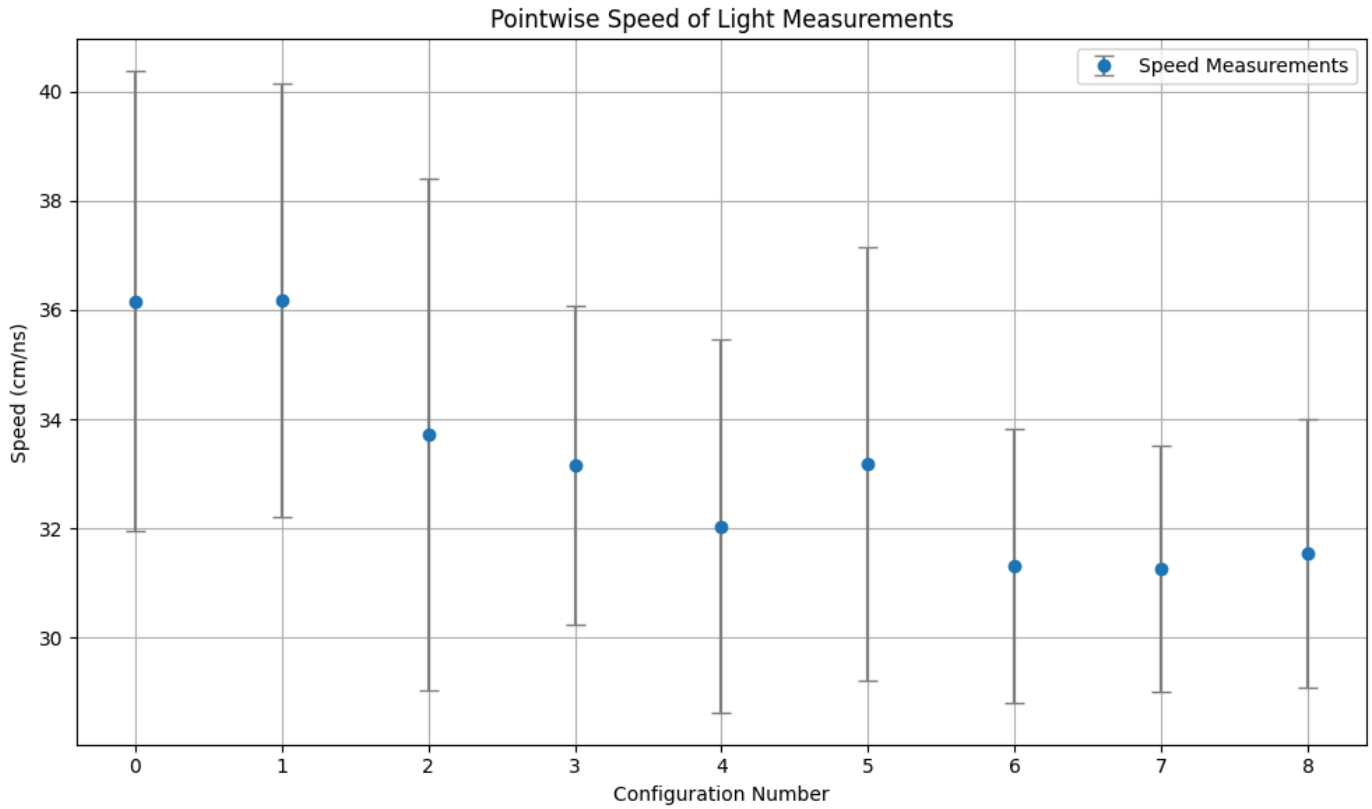


Figure 4 - Pointwise speed of light measurements with uncertainty

Result Evaluation

Using a line of best fit allows us to neglect sources of uncertainty that produce constant global shifts to our data points--for instance, the depth of the sensors in the light emitter. The nature of this method also prevents sensitivity to outliers, though there are no clear outliers in our dataset--that is to say, in the context of the bias-variance tradeoff, linear regression has a much lower variance.

A 1σ confidence interval around our computed value of the speed of light using the linear regression method is:

$$\left(2.58 \cdot 10^8 \frac{\mathrm{m}}{s}, 2.86 \cdot 10^8 \frac{\mathrm{m}}{s}\right)$$

Can stay on same line as text

paragraph form

We find that our computed value is not within 1σ of the accepted value, approximately $2.9 \cdot 10^8 \frac{\mathrm{m}}{\mathrm{s}}$.

A 1σ confidence interval around our pointwise method also does not contain the accepted value.

We also find that the accepted value does not fall within 2σ of either empirical result.

These findings suggest that we have sources of systematic error in our procedure. ~~We believe our~~ estimates of length uncertainty could have been more conservative, for instance. Reflecting on our measurements for the time interval taken by the light pulse, we also believe our methodology produced an interval around a peak that was too wide--resulting in a midpoint that did not properly capture the peak's true center. This is loosely corroborated by our wide time uncertainties, nearing one to two nanoseconds.

Experiment Assumptions and Limitations

talk about distance measurement limitations too

We assume the pulse of light travels at a constant speed across the outward and return journeys, treating the brief changes in medium that occur at the Fresnel lens and the surface of the retroreflector to be negligible and approximating the air's composition as uniform. We also assume the quantity $299,792,458\ \mathrm{m}\ /\ \mathrm{s}$ (~~a value commonly used as the speed of light in a vacuum~~) holds as the speed of light in the laboratory air. One can also view this assumption as: "We assume the laboratory environment to sufficiently approximate a vacuum."

$n_{air} \approx n_{vac}$

yes, this is fine to assume

Discussion and Conclusion

We obtained an emperical value for the speed of light c by comparing a measured distance traveled to a measured time taken. This value does not fall into 1σ of the accepted value, so we conclude our experiment involved sources of uncertainty not adequately accounted for. We believe these are primarily associated with deviations in instrument orientations that cause a light beam to take a longer path than measured, as well as an imprecise approach to obtaining time of exit and return measurements.

& distance measurements

*Restate your key values here

References

[1] Knight, R. D. (2017). *Physics for Scientists and Engineers. Global Edition : a Strategic Approach with Modern Physics*. Pearson.

need to reference this earlier in the text.

make sure to keep same tense throughout.